1.5 Measuring and Constructing Angles

Name _____ Period ____ Date _____

Geometry 1.5 · Student sheet

I can…

- name, measure, and classify angles.
- use the Angle Addition Postulate to find angle measures.
- use an angle bisector to find angle measures.

Vocabulary · fill in the missing word

Angle — a figure formed by two rays with the same _____, called the vertex

Acute angle — an angle that _____ between 0° and 90°

Right angle — an angle that _____ exactly 90°

Obtuse angle — an angle that _____ between 90° and 180°

Angle bisector — a ray that divides an angle into two _____ angles

Part 1 · Discovery: fold an angle in half

Draw ∠ABC on patty paper and measure it. Fold $\overline{BA}$ onto $\overline{BC}$, crease, and draw ray BD along the crease.

m∠ABC	m∠ABD	m∠DBC	m∠ABD + m∠DBC

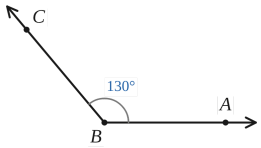
What do you notice?

Part 2 · Notes and examples (fill in each note, then work the example)

Naming angles: ∠ABC, ∠CBA, and ∠B can name the same angle; the vertex _____ is the middle letter.

Example 1

Name the angle in three ways. Then classify it.



Example 2

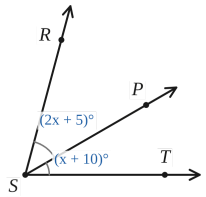
Classify each angle as acute, right, obtuse, or straight.

a. 35° b. 90° c. 152° d. 180°

Angle Addition Postulate: if P is in the interior of ∠RST, then m∠RSP + m∠PST = _____.

Example 3

m∠RST = 75°. Find x, m∠RSP, and m∠PST.



Angle bisector: it divides an angle into two _____ angles, so each half is half the whole.

Example 4

Ray YW bisects ∠XYZ. m∠XYW = (4x − 3)° and m∠WYZ = (2x + 11)°. Find x and m∠XYZ.

Example 5

The peak of a roof makes a 130° angle. A support beam bisects the angle. What angle does the beam make with each side of the roof?

Part 3 · Partner practice

Partner 1

D is in the interior of ∠ABC. m∠ABD = 38° and m∠DBC = 52°. Find m∠ABC and classify it.

Partner 2

A ray bisects an angle into ∠1 and ∠2. m∠1 = (5x − 4)° and m∠2 = (3x + 12)°. Find x and the measure of the whole angle.